# Untrusted input boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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WORKBOOK · 1 OF 15

## Recall and choose your starting point

> Record your starting explanation for Untrusted input boundaries before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### External side effects and recovery

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Approval gates and idempotency

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 15

## Practise with fading help

> Complete the supported Untrusted input boundaries trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Untrusted input** boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

## Individual reasoning check

> Check the proposed Untrusted input boundaries explanation against the supplied facts before discussing it.

Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 15

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Changed independent case

> **Input** format is exactly one ASCII uppercase letter followed by one digit, with no trimming. Evaluate C0, c0, C00, "C0 ", and None. Does a valid format grant access?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 15

## W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Untrusted input boundaries facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

WORKBOOK · 6 OF 15

## W2 Trace the worked case

> Trace the worked case. Apply the supplied Untrusted input boundaries facts and keep your reasoning.

**Input**s A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

WORKBOOK · 7 OF 15

## W3 Audit the limit

> Audit the limit. Apply the supplied Untrusted input boundaries facts and keep your reasoning.

A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

WORKBOOK · 8 OF 15

## W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Untrusted input boundaries facts and keep your reasoning.

Classify B2, b2, B22 and B2; DROP under the exact format.

**My reasoning and evidence**

WORKBOOK · 9 OF 15

## W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Untrusted input boundaries facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

WORKBOOK · 10 OF 15

## Project checkpoint and reflection

> Connect Untrusted input boundaries to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **untrusted input** boundaries **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **untrusted input** boundaries.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 12 OF 15**

## Pictorial case practice: Untrusted input boundaries

> Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one digit, with no trimming. Explain the labelled step before checking the changed condition.

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one digit, with no trimming.

**Input**s h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

AI ACADEMY / AI-6/18

# Untrusted input boundaries

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Validate h3

Two characters with the required case and digit pass.



## 2. Reject wrong shapes

H3 has wrong case; h30 and the trailing-space input have wrong length.



## 3. Check authority next

An accepted identifier still needs trusted authorisation.

**CHANGE ONE THING**

Replace h3 with z0.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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WORKBOOK · 13 OF 15

## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 14 OF 15

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one digit, with no trimming.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? An accepted identifier still needs trusted authorisation.

> 1. Reject wrong shapes

2. Check authority next

3. Validate h3

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Check authority next An accepted identifier still needs trusted authorisation. **Input** shape and permission answer different questions.

Silently normalising these **input**s would change the stated contract.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Replace h3 with z0. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The shape passes; resource access is still undecided.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: The shape passes; resource access is still undecided. Structure contains no **evidence** of ownership or user permission.

**Repair a missing building block**

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Approval gates and idempotency (ai-6/15). Goal: Explain the **mechanism** and terminology of approval gates and idempotency. Short practice: Trace the two deliveries of approved K9.

How many reservations should occur, and why can a new K10 for desk 5 not borrow K9's approval?. Return to this week and explain what you changed.

Review Logs traces and privacy (ai-6/12). Goal: Explain the **mechanism** and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace.

Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each Untrusted input boundaries picture from its labels, then record a relevant limit.

### 01. Untrusted input boundaries: the case



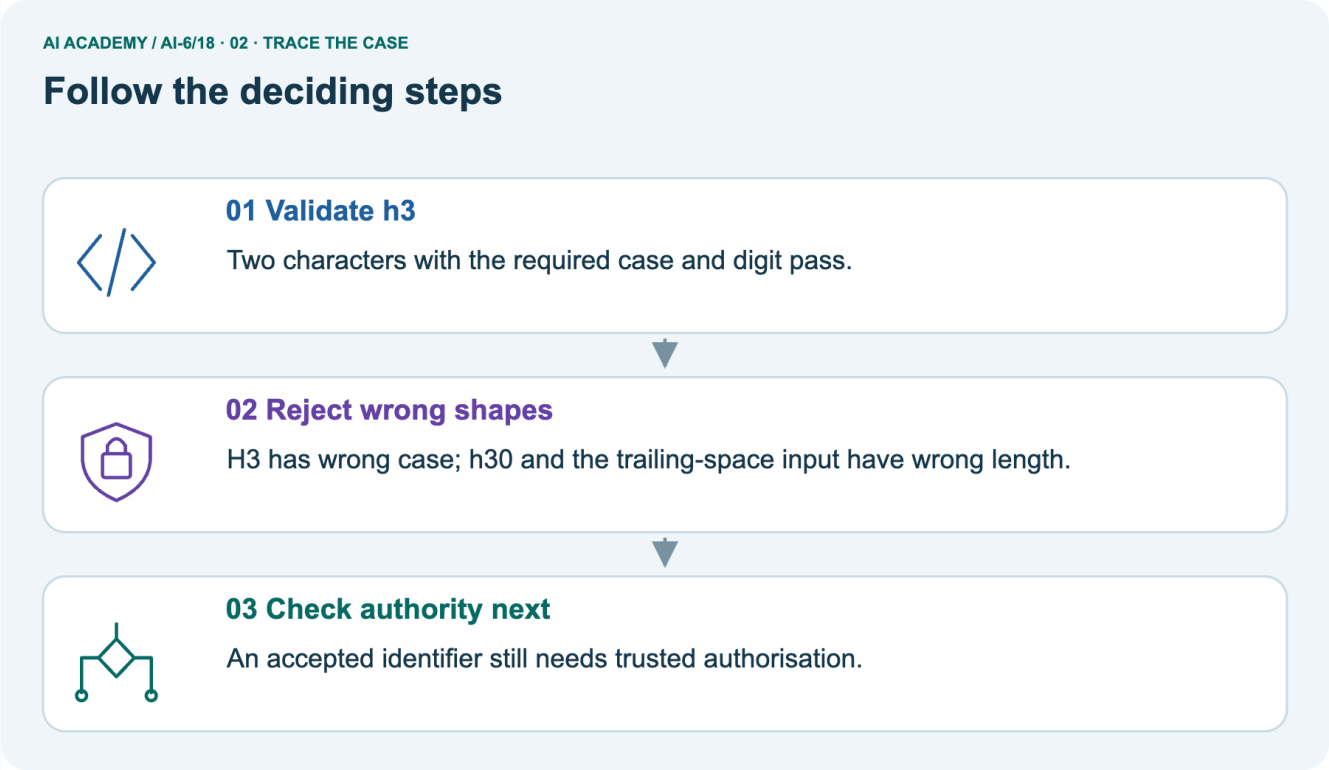

Follow the labelled picture with the matching explanation. Open the vector diagram

Check shape before permission. A toy tool accepts exactly one ASCII lowercase letter followed by one digit, with no trimming. Supplied facts: **Input**s h3, H3, h30 and "h3 ". Shape acceptance never grants access by itself.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



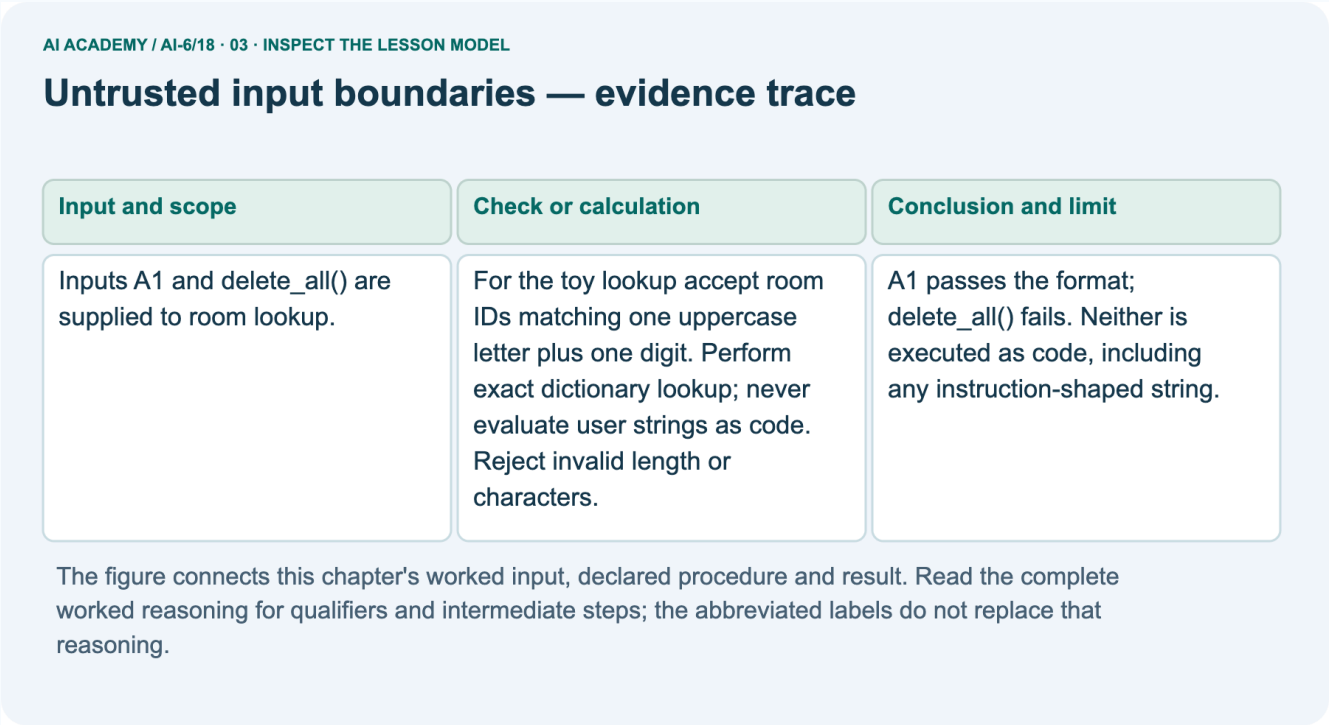

Follow the labelled picture with the matching explanation. Open the vector diagram

Validate h3: Two characters with the required case and digit pass. → Reject wrong shapes: H3 has wrong case; h30 and the trailing-space **input** have wrong length. → Check authority next: An accepted identifier still needs trusted authorisation.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Untrusted input boundaries — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

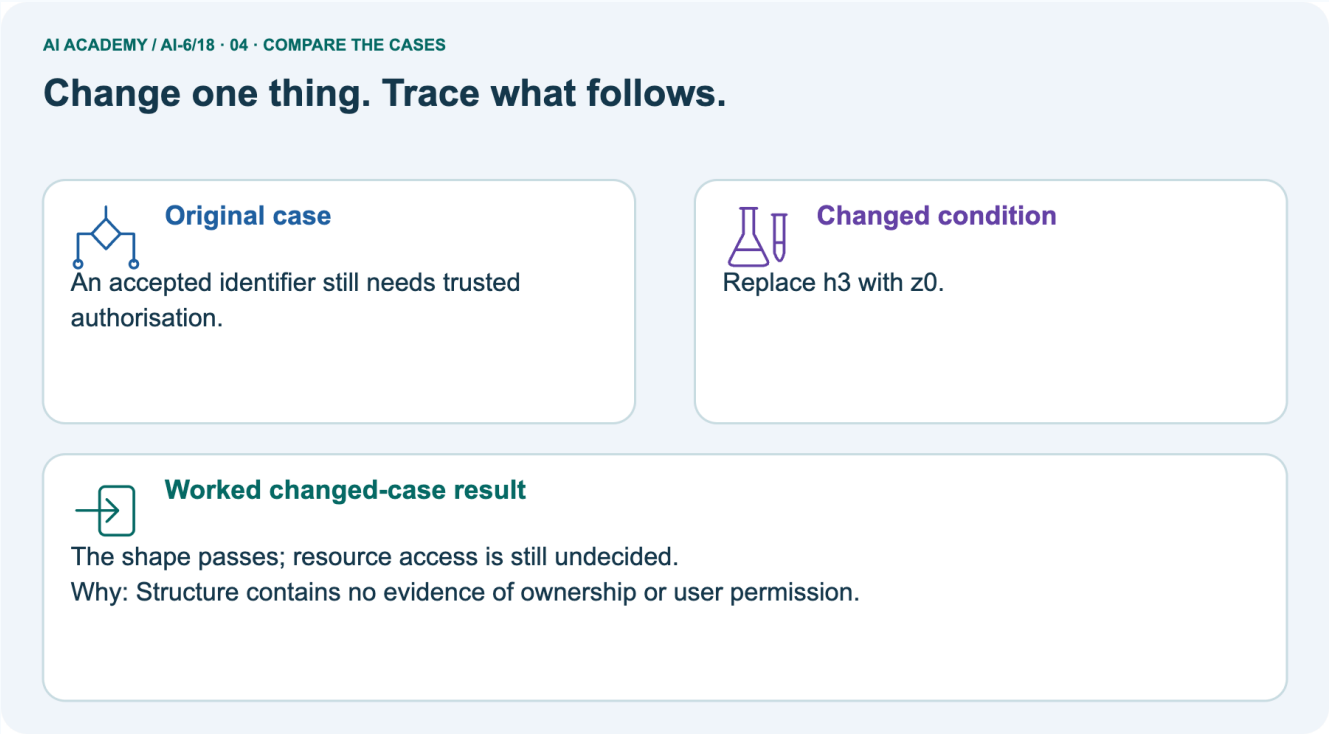
**Untrusted input** boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

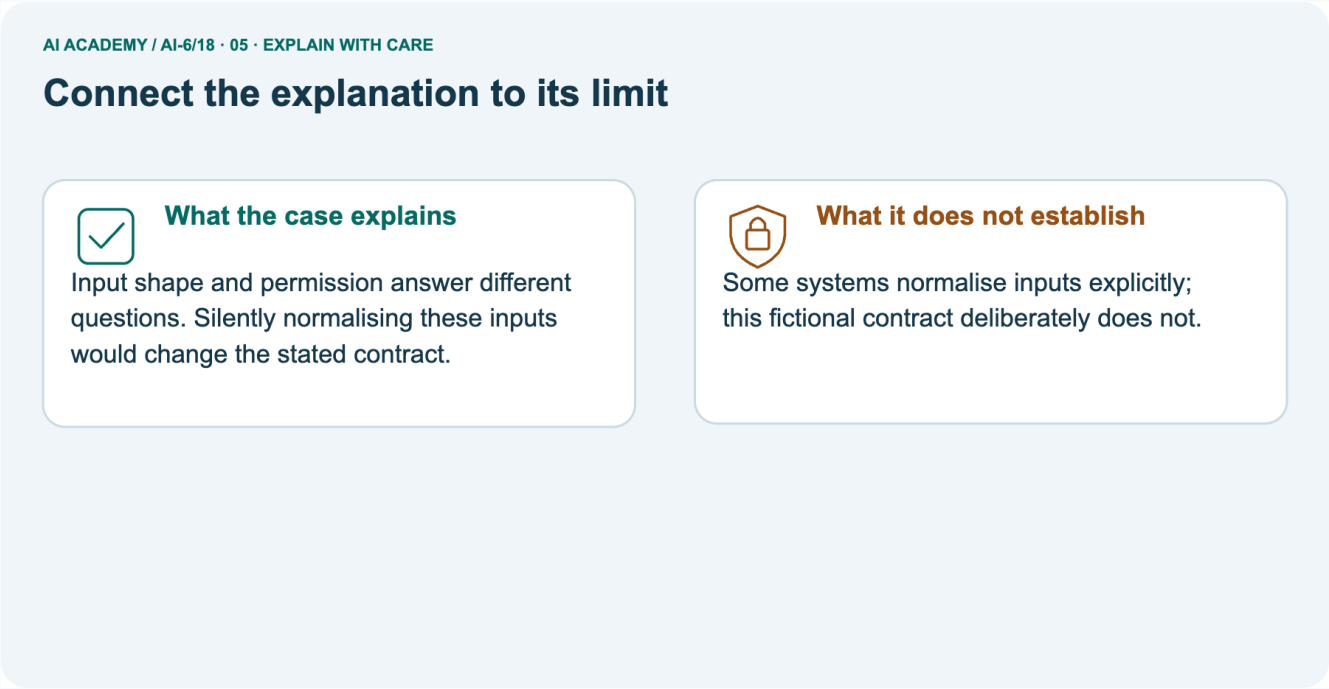
Original case: An accepted identifier still needs trusted authorisation. Changed condition: Replace h3 with z0. Expected result: The shape passes; resource access is still undecided.

Reason: Structure contains no **evidence** of ownership or user permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit



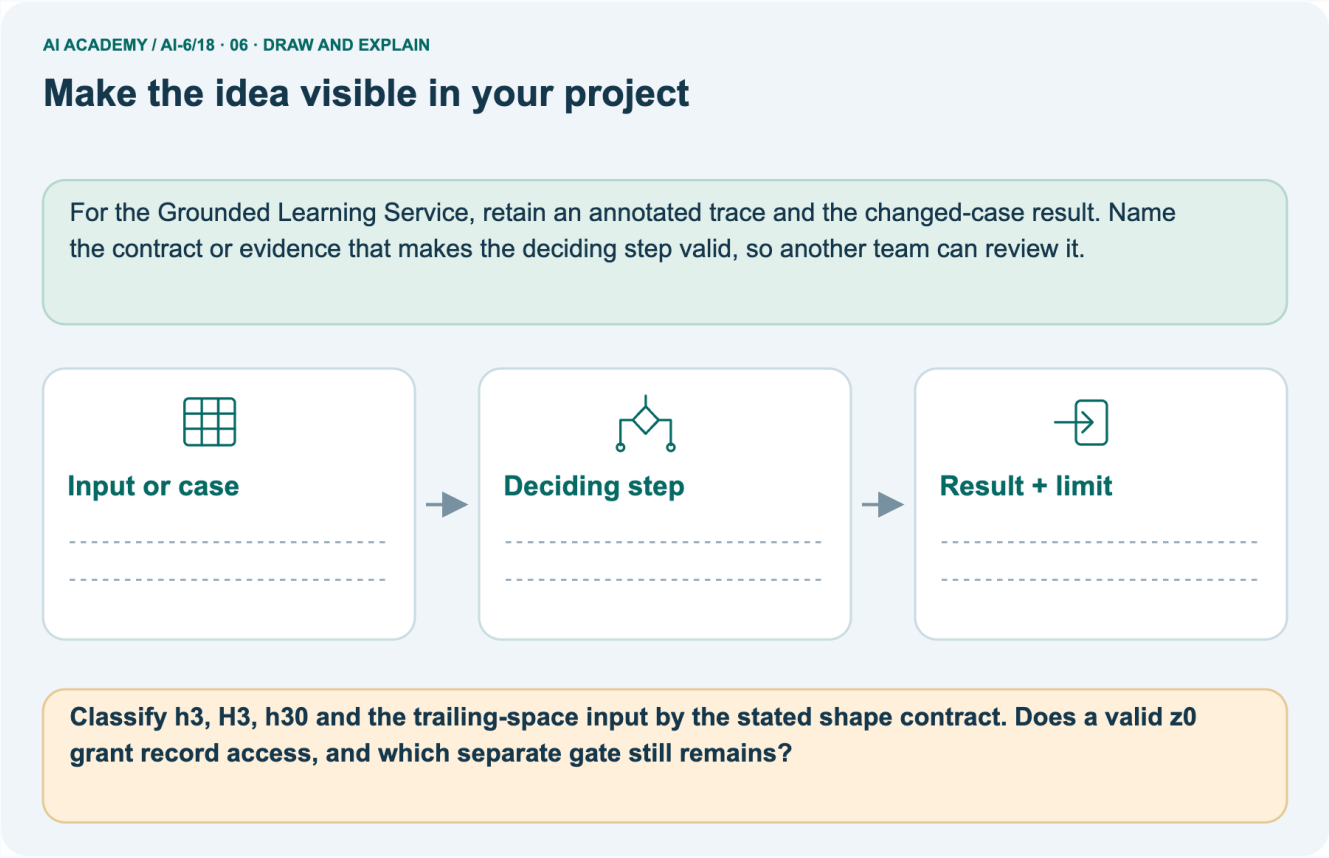

Follow the labelled picture with the matching explanation. Open the vector diagram

**Input** shape and permission answer different questions. Silently normalising these **input**s would change the stated contract. Limit: Some systems normalise **input**s explicitly; this fictional contract deliberately does not.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract. Does a valid z0 grant record access, and which separate gate still remains?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.